

SOLIDWORKS API (NET VBA C#) — Reference (264/337): ISketch (part 2 of 3)

SOLIDWORKS API Documentation (converted from CHM)

ISketch (part 2 of 3) (Type)

SOLIDWORKS API Help

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GetSplineCount Method (ISketch)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISketch Interface](#) : GetSplineCount Method (ISketch)

Gets the number of splines in this sketch.

.NET Syntax

Visual Basic (Declaration)

```
Function GetSplineCount( _  
    ByRef PointCount As System.Integer _  
) As System.Integer
```

Visual Basic (Usage)

```
Dim instance As ISketch  
Dim PointCount As System.Integer  
Dim value As System.Integer  
  
value = instance.GetSplineCount(PointCount)
```

C#

```
System.int GetSplineCount(  
    ref System.int PointCount  
)
```

C++/CLI

```
System.int GetSplineCount(  
& System.int% PointCount  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

PointCount
Number of points in this sketch

Return Value

Number of splines in the sketch

Visual Basic for Applications (VBA) Syntax

See [Sketch::GetSplineCount](#).

Example

[Get Each Spline's Parameters \(C#\)](#)
[Get Each Spline's Parameters \(VB.NET\)](#)
[Get Each Spline's Parameters \(VBA\)](#)

Remarks

Call this method before calling [ISketch::IGetSplines](#) to get the size of the array for that method.

See Also

[ISketch Interface](#)
[ISketch Members](#)
[ISketch::GetSplines Method](#)
[ISketch::GetSplineParamsCount2 Method](#)
[ISketch::GetSplinesInterpolate Method](#)
[ISketch::GetSplineParams2 Method](#)

GetSplineInterpolateCount Method (ISketch)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISketch Interface](#) : GetSplineInterpolateCount Method (ISketch)

Gets the number of points in the spline and number of splines in the sketch.

.NET Syntax

Visual Basic (Declaration)

```
Function GetSplineInterpolateCount( _  
    ByRef PointCount As System.Integer _  
) As System.Integer
```

Visual Basic (Usage)



```
Dim instance As ISketch  
Dim PointCount As System.Integer  
Dim value As System.Integer  
  
value = instance.GetSplineInterpolateCount(PointCount)
```

C#

```
System.int GetSplineInterpolateCount(  
    ref System.int PointCount  
)
```

C++/CLI

```
System.int GetSplineInterpolateCount(  
& System.int% PointCount  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

PointCount

Number of points in this sketch

Return Value

Number of splines in this sketch

Visual Basic for Applications (VBA) Syntax

See [Sketch::GetSplineInterpolateCount](#).

Remarks

Call this method before calling [ISketch::IGetSplinesInterpolate](#) to get the size of the array for that method.

See Also

[ISketch Interface](#)
[ISketch Members](#)
[ISketch::GetSplinesInterpolate Method](#)
[ISketch::GetSplineCount Method](#)
[ISketch::GetSplines Method](#)
[ISketch::GetSplineParamsCount2 Method](#)
[ISketch::GetSplineParams2 Method](#)
[ISketch::IGetSplineParams2 Method](#)
[ISketch::IGetSplines Method](#)

GetSplineParams Method (ISketch)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISketch Interface](#) : GetSplineParams Method (ISketch)

Obsolete. Superseded by [ISketch::GetSplineParams2](#) and [ISketch::IGetSplineParams2](#).

.NET Syntax

Visual Basic (Declaration)

```
Function GetSplineParams() As System.Object
```

Visual Basic (Usage)



```
Dim instance As ISketch  
Dim value As System.Object  
  
value = instance.GetSplineParams()
```

C#

```
System.object GetSplineParams()
```

C++/CLI

```
System.Object^ GetSplineParams();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Visual Basic for Applications (VBA) Syntax

See [Sketch::GetSplineParams](#).

See Also

[ISketch Interface](#)
[ISketch Members](#)

SOLIDWORKS API Help

[Send comments](#) on this topic.

GetSplineParams2 Method (ISketch)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISketch Interface](#) : GetSplineParams2 Method (ISketch)

Obsolete. Superseded by [ISketch::GetSplineParams3](#).

.NET Syntax

Visual Basic (Declaration)

```
Function GetSplineParams2() As System.Object
```

Visual Basic (Usage)



```
Dim instance As ISketch  
Dim value As System.Object  
  
value = instance.GetSplineParams2()
```

C#

```
System.object GetSplineParams2()
```

C++/CLI

```
System.Object^ GetSplineParams2();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Array of doubles (see **Remarks**)

Visual Basic for Applications (VBA) Syntax

See [Sketch::GetSplineParams2](#).

Remarks

See [ISketch::GetSketchSegments](#) or [ISketch::IEnumSketchSegments](#) for access to individual [ISketchSegment](#) and [ISketchSpline](#) objects.

The return value is an array of doubles containing data for all the splines in the sketch:

[*packedDouble1*, *packedDouble2*, *ControlPoint1[Dimension elements]*, *ControlPoint2[Dimension elements]*,... *knot1*, *knot2*,..., *packedDouble3*, *packedDouble4*, *packedDouble5*,]

packedDouble1* and *packedDouble2

The first two array elements for each spline contain four integer values holding information that describes the rest of the data in that spline's parameters:

Spline Element	Packed Data	
	low part	high part
0	<i>Dim</i>	<i>Order</i>
1	<i>nCtrlPoints</i>	<i>Periodic</i>

where:

- *Dim* is the number of dimensions in which the spline is defined
- *Order* is the order of the spline
- *nCtrlPoints* is the number of control points
- *Periodic* is 1 for a closed spline or 0 for an open spline

NOTE: For information about unpacking double arrays into integer pairs, see:

- [Unpacking Double Arrays into Integer Pairs in VB.NET and Visual Basic](#)
- [Unpacking Double Arrays into Integer Pairs in C++](#)
- [Unpacking Double Arrays into Integer Pairs in C#](#)

ControlPoint#

The ControlPoint data (in the sketch coordinate system) follows the two packed data elements.

knots#

The number of knots depends on whether the spline is periodic or not:

Periodic:	$\text{numKnots} = \text{nCtrlPoints} + 1$
Non-Periodic:	$\text{numKnots} = \text{nCtrlPoints} + \text{Order}$

packedDouble3*, *packedDouble4*, and *packedDouble5

The last three array elements for each spline contain five integer values holding style and layer information:

Spline Element	Packed Data	
	low part	high part
<i>i</i>	<i>Color</i>	<i>lineStyle</i>
<i>i+1</i>	<i>lineWidth</i>	<i>Layer</i>
<i>i+2</i>	<i>layerOverride</i>	<i>Not used</i>

where:

- *i* is the index following the last Knot or [2 + numKnots + numControlPointDoubles * Dim]
- *Color* is the COLORREF value describing the color used for the *ith* spline
- *lineStyle* is the line style used for the *ith* spline. Valid values can be found in the [swLineStyles_e](#) enumeration
- *lineWidth* is line width used for the *ith* spline. Valid values can be found in the [swLineWeights_e](#) enumeration
- *Layer* is an integer index to the layer that the *ith* spline is on
- *layerOverride* is integer with bit flags set to determine which properties, if any, have been overridden or should be overridden.

Therefore, the size of the data for each spline is given by:

$$2 + \text{numKnots} + \text{numControlPointDoubles} * \text{Dim} + 3$$

See Also

[ISketch Interface](#)
[ISketch Members](#)
[ISketch::IGetSplineParams2 Method](#)
[ISketch::GetSplineCount Method](#)
[ISketch::GetSplineInterpolateCount Method](#)
[ISketch::GetSplines Method](#)
[ISketch::GetSplinesInterpolate Method](#)
[ISketch::IGetSplines Method](#)
[ISketch::IGetSplinesInterpolate Method](#)

Availability

SOLIDWORKS 2000 FCS, Revision Number 8.0

SOLIDWORKS API Help

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GetSplineParams3 Method (ISketch)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISketch Interface](#) : GetSplineParams3 Method (ISketch)

Obsolete. Superseded by [ISketch::GetSplineParams4](#).

.NET Syntax

Visual Basic (Declaration)

```
Function GetSplineParams3( _  
    ByVal ForceNonPeriodic As System.Boolean _  
) As System.Object
```

Visual Basic (Usage)

```
Dim instance As ISketch  
Dim ForceNonPeriodic As System.Boolean  
Dim value As System.Object  
  
value = instance.GetSplineParams3(ForceNonPeriodic)
```

C#

```
System.object GetSplineParams3(  
    System.bool ForceNonPeriodic  
)
```

C++/CLI

```
System.Object^ GetSplineParams3(  
    & System.bool ForceNonPeriodic  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

ForceNonPeriodic

True to attempt to convert all non-periodic splines to periodic, false to not

Return Value

Array of doubles (see [Remarks](#))

Visual Basic for Applications (VBA) Syntax

See [Sketch::GetSplineParams3](#).

Example

[Get Spline's Parameters \(C#\)](#)
[Get Spline's Parameters \(VB.NET\)](#)
[Get Spline's Parameters \(VBA\)](#)

Remarks

See [ISketch::GetSketchSegments](#) or [ISketch::!EnumSketchSegments](#) for access to individual [ISketchSegment](#) and [ISketchSpline](#) objects.

The return value is an array of doubles containing data for all the splines in the sketch:

[*packedDouble1*, *packedDouble2*, *ControlPoint1*[*Dimension elements*], *ControlPoint2*[*Dimension elements*],... *knot1*, *knot2*,..., *packedDouble3*, *packedDouble4*, *packedDouble5*,]

packedDouble1 and *packedDouble2*

The first two array elements for each spline contain four integer values holding information that describes the rest of the data in that spline's parameters:

Spline Element	Packed Data	
	low part	high part
0	<i>Dim</i>	<i>Order</i>
1	<i>nCtrlPoints</i>	<i>Periodic</i>

where:

- *Dim* is the number of dimensions in which the spline is defined
- *Order* is the order of the spline
- *nCtrlPoints* is the number of control points
- *Periodic* is 1 for a closed spline or 0 for an open spline

NOTE: For information about unpacking double arrays into integer pairs, see:

- [Unpacking Double Arrays into Integer Pairs in VB.NET and Visual Basic](#)
- [Unpacking Double Arrays into Integer Pairs in C++](#)
- [Unpacking Double Arrays into Integer Pairs in C#](#)

ControlPoint#

The ControlPoint data (in the sketch coordinate system) follows the two packed data elements.

knots#

The number of knots depends on whether the spline is periodic or not:

Periodic: numKnots = nCtrlPoints + 1

Non-Periodic: numKnots = nCtrlPoints + Order

packedDouble3, *packedDouble4*, and *packedDouble5*

The last three array elements for each spline contain five integer values holding style and layer information:

Spline Element	Packed Data
----------------	-------------

	low part	high part
i	<i>Color</i>	<i>lineStyle</i>
i+1	<i>lineWidth</i>	<i>Layer</i>
i+2	<i>layerOverride</i>	<i>Not used</i>

where:

- i is the index following the last Knot or $[2 + \text{numKnots} + \text{numControlPointDoubles} * \text{Dim}]$
- *Color* is the COLORREF value describing the color used for the *ith* spline
- *lineStyle* is the line style used for the *ith* spline. Valid values can be found in the [swLineStyle_e](#) enumeration
- *lineWidth* is line width used for the *ith* spline. Valid values can be found in the [swLineWeights_e](#) enumeration
- *Layer* is an integer index to the layer that the *ith* spline is on
- *layerOverride* is integer with bit flags set to determine which properties, if any, have been overridden or should be overridden.

Therefore, the size of the data for each spline is given by:

$$2 + \text{numKnots} + \text{numControlPointDoubles} * \text{Dim} + 3$$

See Also

[ISketch Interface](#)
[ISketch Members](#)
[ISketch::IGetSplineParams3 Method](#)
[ISketch::IGetSplineParamsCount3 Method](#)
[ISketch::IGetSplines Method](#)
[ISketch::IGetSplines Method](#)
[ISketch::IGetSplinesInterpolate Method](#)
[ISketch::IGetSplinesInterpolate Method](#)
[ISketch::IGetSplineCount Method](#)
[ISketch::IGetSplineInterpolateCount Method](#)

Availability

SOLIDWORKS 2010 FCS, Revision Number 18.0

SOLIDWORKS API Help

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GetSplineParams4 Method (ISketch)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISketch Interface](#) : GetSplineParams4 Method (ISketch)

Obsolete. Superseded by [ISketch::GetSplineParams5](#).

.NET Syntax

Visual Basic (Declaration)

```
Function GetSplineParams4( _
    ByVal ForceNonPeriodic As System.Boolean _
) As SplineParamData
```

Visual Basic (Usage)



```
Dim instance As ISketch
Dim ForceNonPeriodic As System.Boolean
Dim value As SplineParamData

value = instance.GetSplineParams4(ForceNonPeriodic)
```

C#

```
SplineParamData GetSplineParams4(
    System.bool ForceNonPeriodic
)
```

C++/CLI

```
SplineParamData^ GetSplineParams4(  
& System.bool ForceNonPeriodic  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

ForceNonPeriodic

True to attempt to convert a non-periodic spline to a periodic one, false to not

Return Value

[ISplineParamData](#)

Visual Basic for Applications (VBA) Syntax

See [Sketch::GetSplineParams4](#).

Example

[Get Spline's Parameters \(C#\)](#)
[Get Spline's Parameters \(VB.NET\)](#)
[Get Spline's Parameters \(VBA\)](#)

See Also

[ISketch Interface](#)
[ISketch Members](#)
[ISketch::GetSplineParamsCount3 Method](#)
[ISketch::GetSplines Method](#)
[ISketch::ISplines Method](#)
[ISketch::GetSplinesInterpolate Method](#)
[ISketch::ISplinesInterpolate Method](#)
[ISketch::GetSplineCount Method](#)
[ISketch::GetSplineInterpolateCount Method](#)

Availability

SOLIDWORKS 2011 FCS, Revision Number 19.0

SOLIDWORKS API Help

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GetSplineParams5 Method (ISketch)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISketch Interface](#) : GetSplineParams5 Method (ISketch)

Gets the parameterization data of the specified spline in this sketch.

.NET Syntax

Visual Basic (Declaration)

```
Function GetSplineParams5( _  
    ByVal ForceNonPeriodic As System.Boolean, _  
    ByVal Index As System.Integer _  
) As System.Object
```

Visual Basic (Usage)



```
Dim instance As ISketch  
Dim ForceNonPeriodic As System.Boolean  
Dim Index As System.Integer  
Dim value As System.Object
```

```
value = instance.GetSplineParams5(ForceNonPeriodic, Index)
```

C#

```
System.object GetSplineParams5(  
    System.bool ForceNonPeriodic,
```

```
System.int Index  
)
```

C++/CLI

```
System.Object^ GetSplineParams5(  
& System.bool ForceNonPeriodic,  
& System.int Index  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

ForceNonPeriodic

True to attempt to convert a non-periodic spline to a periodic one, false to not

Index

0-based index indicating the spline whose parameterization data to get

Return Value

[ISplineParamData](#)

Visual Basic for Applications (VBA) Syntax

See [Sketch::GetSplineParams5](#).

Example

[Get Each Spline's Parameters \(C#\)](#)

[Get Each Spline's Parameters \(VB.NET\)](#)

[Get Each Spline's Parameters \(VBA\)](#)

See Also

[ISketch Interface](#)

[ISketch Members](#)

[ISketch::GetSplineParamsCount Method \(\)](#)

[ISketch::GetSplines Method \(\)](#)

[ISketch::IGetSplines Method \(\)](#)

[ISketch::GetSplinesInterpolate Method \(\)](#)

[ISketch::GetSplineInterpolateCount Method \(\)](#)

[ISketch::GetSplineCount Method \(\)](#)

[ISketch::IGetSplinesInterpolate Method \(\)](#)

Availability

SOLIDWORKS 2017 FCS, Revision Number 25.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

GetSplineParamsCount Method (ISketch)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISketch Interface](#) : GetSplineParamsCount Method (ISketch)

Obsolete. Superseded by [ISketch::GetSplineParamsCount2](#).

.NET Syntax

Visual Basic (Declaration)

```
Function GetSplineParamsCount(  
    ByRef Size As System.Integer _  
) As System.Integer
```

Visual Basic (Usage)



```
Dim instance As ISketch  
Dim Size As System.Integer  
Dim value As System.Integer
```

```
value = instance.GetSplineParamsCount(Size)
```

C#

```
System.int GetSplineParamsCount(  
    out System.int Size  
)
```

C++/CLI

```
System.int GetSplineParamsCount(  
& [Out] System.int Size  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Size

Visual Basic for Applications (VBA) Syntax

See [Sketch::GetSplineParamsCount](#).

See Also

[ISketch Interface](#)
[ISketch Members](#)

SOLIDWORKS API Help

[Send comments](#) on this topic.

GetSplineParamsCount2 Method (ISketch)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISketch Interface](#) : GetSplineParamsCount2 Method (ISketch)

Obsolete. Superseded by [ISketch::GetSplineParamsCount3](#).

.NET Syntax

Visual Basic (Declaration)

```
Function GetSplineParamsCount2( _  
    ByRef Size As System.Integer _  
) As System.Integer
```

Visual Basic (Usage)



```
Dim instance As ISketch  
Dim Size As System.Integer  
Dim value As System.Integer  
  
value = instance.GetSplineParamsCount2(Size)
```

C#

```
System.int GetSplineParamsCount2(  
    out System.int Size  
)
```

C++/CLI

```
System.int GetSplineParamsCount2(  
& [Out] System.int Size  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Size

Size of array required for a call to [ISketch::ISketchParams2](#)

Return Value

Number of [splines](#)

Visual Basic for Applications (VBA) Syntax

See [Sketch::GetSplineParamsCount2](#).

See Also

[ISketch Interface](#)
[ISketch Members](#)
[ISketch::GetSplineCount Method](#)
[ISketch::GetSplineInterpolateCount Method](#)
[ISketch::GetSplines Method](#)
[ISketch::GetSplinesInterpolate Method](#)
[ISketch::IGetSplines Method](#)
[ISketch::IGetSplinesInterpolate Method](#)
[ISketch::GetSplineParams2 Method](#)

Availability

SOLIDWORKS 2000 FCS, Revision Number 8.0

SOLIDWORKS API Help

[Send comments](#) on this topic.

GetSplineParamsCount3 Method (ISketch)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISketch Interface](#) : GetSplineParamsCount3 Method (ISketch)

Gets the number of splines in the sketch and the size of array required to hold the data for them.

.NET Syntax

Visual Basic (Declaration)

```
Function GetSplineParamsCount3(  
    ByVal ForceNonPeriodic As System.Boolean, _  
    ByRef Size As System.Integer _  
) As System.Integer
```

Visual Basic (Usage)



```
Dim instance As ISketch  
Dim ForceNonPeriodic As System.Boolean  
Dim Size As System.Integer  
Dim value As System.Integer  
  
value = instance.GetSplineParamsCount3(ForceNonPeriodic, Size)
```

C#

```
System.int GetSplineParamsCount3(  
    System.bool ForceNonPeriodic,  
    out System.int Size  
)
```

C++/CLI

```
System.int GetSplineParamsCount3(  
& System.bool ForceNonPeriodic,  
& [Out] System.int Size  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

ForceNonPeriodic

True to attempt to convert all non-periodic splines to periodic, false to not

Size

Size of array required for a call to [ISketch::IGetSplineParams3](#)

Return Value

Number of [splines](#)

Visual Basic for Applications (VBA) Syntax

See [Sketch::GetSplineParamsCount3](#).

Example

[Get Spline's Parameters \(C#\)](#)
[Get Spline's Parameters \(VB.NET\)](#)
[Get Spline's Parameters \(VBA\)](#)

See Also

[ISketch Interface](#)
[ISketch Members](#)
[ISketch::GetSplines Method](#)
[ISketch::GetSplinesInterpolate Method](#)
[ISketch::IGetSplines Method](#)
[ISketch::IGetSplinesInterpolate Method](#)
[ISketch::GetSplineCount Method](#)
[ISketch::GetSplineInterpolateCount Method](#)
[ISketch::GetSplineParams3 Method](#)

Availability

SOLIDWORKS 2010 FCS, Revision Number 18.0

SOLIDWORKS API Help

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GetSplines Method (ISketch)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISketch Interface](#) : GetSplines Method (ISketch)

Gets information for each spline by tessellation instead of by interpolation as is done by [ISketch::GetSplinesInterpolate](#) and [ISketch::IGetSplinesInterpolate](#).

.NET Syntax

Visual Basic (Declaration)

```
Function GetSplines() As System.Object
```

Visual Basic (Usage)



```
Dim instance As ISketch  
Dim value As System.Object  
  
value = instance.GetSplines()
```

C#

```
System.object GetSplines()
```

C++/CLI

```
System.Object^ GetSplines();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Array of doubles (see [Remarks](#))

Visual Basic for Applications (VBA) Syntax

See [Sketch::GetSplines](#).

Remarks

See [ISketch::GetSketchSegments](#) or [ISketch::IEnumSketchSegments](#) for access to individual [ISketchSegment](#) and [ISketchSpline](#) objects.

The returned array of doubles has the format:

$$\left[\left[\text{Color}, \text{LineType}, \text{NumSplinePoints}, [x, y, z] \right] \right]$$

This complete set of data repeats itself for each spline found in the sketch. For each spline, the array returned contains the color, the line type, the number of spline points in the spline, and the X,Y,Z value for each of those points. Therefore, the $[x, y, z]$ parameter is an array of *NumSplinePoints*, which may vary in size from spline to spline.

The $[x, y, z]$ points for each spline are not the same as the points used to generate the spline. This method tessellates the spline based on the display quality and places points along the spline appropriately. *LineType* may take one of the values in [swLineTypes_e](#).

See Also

[ISketch Interface](#)
[ISketch Members](#)
[ISketch::IGetSplines Method](#)
[ISketch::GetSplineCount Method](#)
[ISketch::GetSplineInterpolateCount Method](#)
[ISketch::GetSplineParams2 Method](#)
[ISketch::GetSplineParamsCount2 Method](#)
[ISketch::IGetSplineParams2 Method](#)
[ISketchSpline Interface](#)

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GetSplinesInterpolate Method (ISketch)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISketch Interface](#) : GetSplinesInterpolate Method (ISketch)

Gets the spline points by interpolation instead of by tessellation as is done by [ISketch::GetSplines](#) and [ISketch::IGetSplines](#).

.NET Syntax

Visual Basic (Declaration)

```
Function GetSplinesInterpolate() As System.Object
```

Visual Basic (Usage)



```
Dim instance As ISketch  
Dim value As System.Object  
  
value = instance.GetSplinesInterpolate()
```

C#

```
System.object GetSplinesInterpolate()
```

C++/CLI

```
System.Object^ GetSplinesInterpolate();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Array of doubles (see [Remarks](#))

Visual Basic for Applications (VBA) Syntax

See [Sketch::GetSplinesInterpolate](#).

Example

[Get Spline Points \(C++\)](#)

Remarks

The return value is an array of doubles formatted as:

$$\left[\left[\text{NumSplinePoints}, [x, y, z] \right], \right]$$

This complete set of data repeats itself for each spline found in the sketch. For each spline, the array returned contains the number of spline points in the spline, and the X,Y,Z value for each of those points.

The $[x,y,z]$ parameter is an array of *NumSplinePoints*. For example, if your sketch has two splines and each spline has three points, then the data would be in the following format:

[3, x1_1, y1_1, z1_1, x2_1, y2_1, z2_1, x3_1, y3_1, z3_1, 3, x1_2, y1_2, z1_2, x2_2, y2_2, z2_2, x3_2, y3_2, z3_2]

The $[x,y,z]$ points for each spline are the same points used to generate the spline.

See Also

[ISketch Interface](#)
[ISketch Members](#)
[ISketch::GetSplineCount Method](#)
[ISketch::GetSplineParams2 Method](#)
[ISketch::GetSplineParamsCount2 Method](#)
[ISketch::IGetSplineParams2 Method](#)
[ISketch::IGetSplinesInterpolate Method](#)
[ISketch::GetSplineInterpolateCount Method](#)

SOLIDWORKS API Help

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GetUserPoints Method (ISketch)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISketch Interface](#) : GetUserPoints Method (ISketch)

Obsolete. Superseded by [ISketch::GetUserPoints2](#) and [ISketch::IGetUserPoints2](#).

.NET Syntax

Visual Basic (Declaration)

```
Function GetUserPoints() As System.Object
```

Visual Basic (Usage)



```
Dim instance As ISketch  
Dim value As System.Object  
  
value = instance.GetUserPoints()
```

C#

```
System.object GetUserPoints()
```

C++/CLI

```
System.Object^ GetUserPoints();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Visual Basic for Applications (VBA) Syntax

See [Sketch::GetUserPoints](#).

See Also

[ISketch Interface](#)
[ISketch Members](#)

SOLIDWORKS API Help

[Send comments on this topic.](#)

GetUserPoints2 Method (ISketch)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISketch Interface](#) : GetUserPoints2 Method (ISketch)

Gets all of the user points in this sketch.

.NET Syntax

Visual Basic (Declaration)

```
Function GetUserPoints2() As System.Object
```

```
Dim instance As ISketch
Dim value As System.Object

value = instance.GetUserPoints2()
```

C#

```
System.Object GetUserPoints2()
```

C++/CLI

```
System.Object^ GetUserPoints2();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Array of doubles (see **Remarks**)

Visual Basic for Applications (VBA) Syntax

See [Sketch::GetUserPoints2](#).

Example

[Get Sketch Points and Their Persistent IDs \(VBA\)](#)

Remarks

The return value is an array of doubles with this format:

[*Color*, *Line Font*, *Line Width*, *Layer ID*, *Layer Override*, *ptLoc*[3]]

where:

- *Color* = COLORREF returned as an integer. Return value could be 0 or -1 for default color.
- *Line Font* = line style. Valid line styles can be found in the [swLineStyles_e](#) enumeration.
- *Line Width* = integer value defining the line width. Valid width values can be found in the [swLineWeights_e](#) enumeration.
- *Layer ID* = integer value indicating which layer holds this entity. The [ILayer](#) object can be obtained by passing this integer value to [ILayerMgr::GetLayerById](#) or [ILayerMgr::!GetLayerById](#).
- *Layer Override* = integer with bit flags set to determine which properties, if any, have been overridden with respect to the Layer default properties. If the bit value is set, then the specific property or properties have been overridden. The bit indicators are: color = 0x1, style : 0x2, and width = 0x4. Therefore, if LayerOverride was returned as 3, you know the color and style have been specifically set for this item and may not match the default values associated with this item's layer.
- *ptLoc*[3] = an array of doubles (x, y, z) describing the point location

This set of data repeats for each user point in the sketch. The size of the array is (NumPts * 8). To determine the number of points in the sketch, use [ISketch::GetUserPointsCount](#).

See [ISketch::GetSketchPoints2](#), [ISketch::!GetSketchPoints2](#), or [ISketch::!EnumSketchPoints](#) for access to individual [ISketchPoint](#) objects.

See Also

[ISketch Interface](#)
[ISketch Members](#)
[ISketch::!GetUserPoints2 Method](#)

Availability

SOLIDWORKS 2003 FCS, Revision Number 11.0

GetUserPointsCount Method (ISketch)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISketch Interface](#) : GetUserPointsCount Method (ISketch)

Gets the number of user points in the sketch.

.NET Syntax

Visual Basic (Declaration)

```
Function GetUserPointsCount() As System.Integer
```

Visual Basic (Usage)



```
Dim instance As ISketch  
Dim value As System.Integer  
  
value = instance.GetUserPointsCount()
```

C#

```
System.int GetUserPointsCount()
```

C++/CLI

```
System.int GetUserPointsCount();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Number of user points in the sketch

Visual Basic for Applications (VBA) Syntax

See [Sketch::GetUserPointsCount](#).

Example

[Get Polylines Information \(VBA\)](#)

Remarks

Call this method before calling [ISketch::IGetUserPoints2](#) to get the size of the array for that method.

See Also

[ISketch Interface](#)
[ISketch Members](#)
[ISketch::IGetUserPoints2 Method](#)
[ISketch::GetUserPoints2 Method](#)
[ISketch::GetUserPointsCount Method](#)
[ISketch::IGetUserPoints2 Method](#)

SOLIDWORKS API Help

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IEnumSketchHatches Method (ISketch)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISketch Interface](#) : IEnumSketchHatches Method (ISketch)

Gets the sketch hatches enumeration in this sketch.

.NET Syntax

Visual Basic (Declaration)

```
Function IEnumSketchHatches() As EnumSketchHatches
```

Visual Basic (Usage)



```
Dim instance As ISketch  
Dim value As EnumSketchHatches  
  
value = instance.IEnumSketchHatches()
```

C#

```
EnumSketchHatches IEnumSketchHatches()
```

C++/CLI

```
EnumSketchHatches^ IEnumSketchHatches();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

[Sketch hatches enumeration](#)

Visual Basic for Applications (VBA) Syntax

See [Sketch::IEnumSketchHatches](#).

See Also

[ISketch Interface](#)
[ISketch Members](#)
[ISketchHatch::GetSketchHatches Method](#)

Availability

SOLIDWORKS 2000 FCS, Revision Number 8.0

SOLIDWORKS API Help

[Send comments](#) on this topic.

IEnumSketchPoints Method (ISketch)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISketch Interface](#) : IEnumSketchPoints Method (ISketch)

Gets the sketch points enumeration in this sketch.

.NET Syntax

Visual Basic (Declaration)

```
Function IEnumSketchPoints() As EnumSketchPoints
```

Visual Basic (Usage)



```
Dim instance As ISketch  
Dim value As EnumSketchPoints  
  
value = instance.IEnumSketchPoints()
```

C#

```
EnumSketchPoints IEnumSketchPoints()
```

C++/CLI

```
EnumSketchPoints^ IEnumSketchPoints();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

[Sketch points enumeration](#)

Visual Basic for Applications (VBA) Syntax

See [Sketch::IEnumSketchPoints](#).

Example

[Select All Sketch Segments \(C++\)](#)

See Also

[ISketch Interface](#)
[ISketch Members](#)
[ISketch::GetSketchPoints2 Method](#)
[ISketch::GetSketchPointsCount2 Method](#)
[ISketch::GetUserPoints2 Method](#)
[ISketch::GetUserPointsCount Method](#)
[ISketch::IGetUserPoints2 Method](#)

Availability

SOLIDWORKS 99, datecode 1999207

SOLIDWORKS API Help

[Send comments on this topic.](#)

IEnumSketchSegments Method (ISketch)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISketch Interface](#) : IEnumSketchSegments Method (ISketch)

Gets the sketch segments enumeration in this sketch.

.NET Syntax

Visual Basic (Declaration)

```
Function IEnumSketchSegments() As EnumSketchSegments
```

Visual Basic (Usage)



```
Dim instance As ISketch  
Dim value As EnumSketchSegments  
  
value = instance.IEnumSketchSegments()
```

C#

```
EnumSketchSegments IEnumSketchSegments()
```

C++/CLI

```
EnumSketchSegments^ IEnumSketchSegments();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

[Sketch segments enumeration](#)

Visual Basic for Applications (VBA) Syntax

See [Sketch::IEnumSketchSegments](#).

Example

[Select All Sketch Segments \(C++\)](#)

Remarks

Sketch segments include line, arc, spline, parabola, and ellipse entities.

See Also

[ISketch Interface](#)
[ISketch Members](#)
[ISketch::GetSketchSegments Method](#)
[ISketch::GetSketchTextSegments Method](#)
[ISketch::IEnumSketchTextSegments Method](#)

Availability

IEnumSketchTextSegments Method (ISketch)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISketch Interface](#) : IEnumSketchTextSegments Method (ISketch)

Gets the sketch segments enumeration for the selected text in this sketch.

.NET Syntax

Visual Basic (Declaration)

```
Function IEnumSketchTextSegments() As EnumSketchSegments
```

Visual Basic (Usage)



```
Dim instance As ISketch  
Dim value As EnumSketchSegments  
  
value = instance.IEnumSketchTextSegments()
```

C#

```
EnumSketchSegments IEnumSketchTextSegments()
```

C++/CLI

```
EnumSketchSegments^ IEnumSketchTextSegments();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

[Sketch segments enumeration](#)

Visual Basic for Applications (VBA) Syntax

See [Sketch::IEnumSketchTextSegments](#).

See Also

[ISketch Interface](#)
[ISketch Members](#)
[ISketch::GetSketchTextSegments Method](#)
[ISketch::GetSketchSegments Method](#)
[ISketch::IEnumSketchSegments Method](#)

Availability

SOLIDWORKS 2000 FCS, Revision Number 8.0

IGetArcs Method (ISketch)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISketch Interface](#) : IGetArcs Method (ISketch)

Obsolete. Superseded by [ISketch::GetArcs2](#) and [ISketch::IGetArcs2](#).

.NET Syntax

Visual Basic (Declaration)

```
Function IGetArcs() As System.Double
```

Visual Basic (Usage)



```
Dim instance As ISketch  
Dim value As System.Double
```

```
value = instance.IGetArcs()
```

C#

```
System.double IGetArcs()
```

C++/CLI

```
System.double IGetArcs();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Visual Basic for Applications (VBA) Syntax

See [Sketch::IGetArcs](#).

See Also

[ISketch Interface](#)
[ISketch Members](#)

SOLIDWORKS API Help

[Send comments](#) on this topic.

IGetArcs2 Method (ISketch)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISketch Interface](#) : IGetArcs2 Method (ISketch)

Gets all of the arcs in the sketch.

.NET Syntax

Visual Basic (Declaration)

```
Function IGetArcs2( _  
    ByVal ArraySize As System.Integer _  
) As System.Double
```

Visual Basic (Usage)



```
Dim instance As ISketch  
Dim ArraySize As System.Integer  
Dim value As System.Double  
  
value = instance.IGetArcs2(ArraySize)
```

C#

```
System.double IGetArcs2(  
    System.int ArraySize  
)
```

C++/CLI

```
System.double IGetArcs2(  
& System.int ArraySize  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

ArraySize

Number of arcs in the sketch

Return Value

- in-process, unmanaged C++: Pointer to an array of doubles containing the arcs in the sketch (see **Remarks**)
 - VBA, VB.NET, C#, and C++/CLI: Not supported
- See [In-process Methods](#) for details about this type of method.

Remarks

To determine the number of arcs, use [ISketch::GetArcCount](#).

Return value is an array of doubles with the format:

[*Color*, *Type*, *Line Font*, *Line Width*, *Layer ID*, *Layer Override*, [*Start*], [*End*], [*Center*] *Rotation Dir*]

where

- *Color* - COLORREF returned as an integer. Return value could be 0 or -1 for default color.
- *Type* - Line type. Valid returns as defined in [swLineTypes.e](#). A line type is a combination of a line style and line weight.
- *Line Font* - Line style. Valid line styles as defined in [swLineStyles.e](#) enumeration.
- *Line Width* - Integer value defining the line width. Valid width values as defined in [swLineWeights.e](#).
- *Layer ID* - Integer value indicating which layer holds this entity. The [ILayer](#) object can be obtained by passing this integer value to [ILayerMgr::GetLayerById](#) or [ILayerMgr::GetLayerById](#).
- *Layer Override* - Integer with bit flags set to determine which properties, if any, have been overridden with respect to the layer default properties. If the bit value is set, then the specific property or properties have been overridden. The bit indicators are: color = 0x1, style = 0x2, and width = 0x4. Therefore, if Layer Override was returned as 3, the color and style have been specifically set for this item and may not match the default values associated with this item's layer.
- *Start[3]* - An array of 3 doubles (X,Y,Z) describing the start point.
- *End[3]* - An array of 3 doubles (X,Y,Z) describing the end point. If the arc is closed, then End[3] is the same point as the Start.
- *Center[3]* - An array of 3 doubles (X,Y,Z) describing the center point.
- *Rotation Dir* - Rotational direction (CW = -1, CCW = 1)

This set of data repeats for each arc in the sketch. The size of the array is (NumArcs * 16).

See Also

[ISketch Interface](#)
[ISketch Members](#)
[ISketch::GetArcs2 Method](#)
[ISketchArc Interface](#)

Availability

SOLIDWORKS 2003 FCS, Revision Number 11.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

IGetContourEdges Method (ISketch)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISketch Interface](#) : IGetContourEdges Method (ISketch)

Gets the edges for a sketch that has one contour.

.NET Syntax

Visual Basic (Declaration)

```
Function IGetContourEdges( _  
    ByRef Xform As System.Double, _  
    ByVal EdgeCount As System.Integer _  
    ) As Edge
```

Visual Basic (Usage)

```
Dim instance As ISketch  
Dim Xform As System.Double  
Dim EdgeCount As System.Integer  
Dim value As Edge
```

```
value = instance.IGetContourEdges(Xform, EdgeCount)
```

C#



```
Edge IGetContourEdges(  
    ref System.double Xform,  
    System.int EdgeCount  
)
```

C++/CLI

```
Edge^ IGetContourEdges(  
& System.double% Xform,  
& System.int EdgeCount  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Xform
Sketch entities, edges
EdgeCount
Number of edges

Return Value

- in-process, unmanaged C++: Pointer to an array of [edges](#)
 - VBA, VB.NET, C#, and C++/CLI: Not supported
- See [In-process Methods](#) for details about this type of method.

Remarks

This method supports sketches with one contour only. For sketches with multiple contours, use [ISketch::GetSketchContours](#) or [ISketch::IGetSketchContours](#) and then [ISketchContour::GetEdges](#) or [ISketchContour::IGetEdges](#).

To get the number of edges, call [ISketch::GetContourEdgesCount](#) before calling this method.

Specifying the unit transform returns the edges in the space of the sketch.

If your application is expecting the returned edges to be in the context of the model's coordinate system, then xform should be the inverse of the transform returned by [ISketch::ModelToSketchTransform](#).

See Also

[ISketch Interface](#)
[ISketch Members](#)
[ISketch::GetContourEdges Method](#)
[ISketchContour Interface](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments on this topic](#).

IGetEllipses Method (ISketch)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISketch Interface](#) : IGetEllipses Method (ISketch)

Obsolete. Superseded by [ISketch::GetEllipses3](#) and [ISketch::IGetEllipses3](#).

.NET Syntax

Visual Basic (Declaration)

```
Function IGetEllipses() As System.Double
```

Visual Basic (Usage)

```
Dim instance As ISketch  
Dim value As System.Double  
  
value = instance.IGetEllipses()
```

C#



```
System.double IGetEllipses()
```

C++/CLI

```
System.double IGetEllipses();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Visual Basic for Applications (VBA) Syntax

See [Sketch::IGetEllipses](#).

See Also

[ISketch Interface](#)
[ISketch Members](#)

SOLIDWORKS API Help

[Send comments](#) on this topic.

IGetEllipses2 Method (ISketch)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISketch Interface](#) : IGetEllipses2 Method (ISketch)

Obsolete. Superseded by [ISketch::GetEllipses3](#) and [ISketch::IGetEllipses3](#).

.NET Syntax

Visual Basic (Declaration)

```
Function IGetEllipses2() As System.Double
```

Visual Basic (Usage)



```
Dim instance As ISketch  
Dim value As System.Double  
  
value = instance.IGetEllipses2()
```

C#

```
System.double IGetEllipses2()
```

C++/CLI

```
System.double IGetEllipses2();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Visual Basic for Applications (VBA) Syntax

See [Sketch::IGetEllipses2](#).

See Also

[ISketch Interface](#)
[ISketch Members](#)

SOLIDWORKS API Help

[Send comments](#) on this topic.

IGetEllipses3 Method (ISketch)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISketch Interface](#) : IGetEllipses3 Method (ISketch)

Gets all of the ellipses in the sketch.

.NET Syntax

```
Function IGetEllipses3( _
    ByVal ArraySize As System.Integer _
) As System.Double
```



```
Dim instance As ISketch
Dim ArraySize As System.Integer
Dim value As System.Double

value = instance.IGetEllipses3(ArraySize)
```

```
System.double IGetEllipses3(
    System.int ArraySize
)
```

```
System.double IGetEllipses3(
    & System.int ArraySize
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

ArraySize

Number of ellipses in the sketch

Return Value

- in-process, unmanaged C++: Pointer to an array of doubles (see **Remarks**)
 - VBA, VB.NET, C#, and C++/CLI: Not supported
- See [In-process Methods](#) for details about this type of method.

Remarks

See [ISketch::GetSketchSegments](#) or [ISketch::IEnumSketchSegments](#) for access to individual [ISketchSegment](#) and [ISketchEllipse](#) objects.

The return values are in an array of doubles:

[*Color*, *LineType*, *LineFont*, *LineWidth*, *LayerID*, *LayerOverride*, *StartPt*[3], *EndPt*[3], *CenterPt*[3], *MajorPt*[3], *MinorPt*[3], *Direction* ...]

where:

<i>Color</i> :	COLORREF returned as an integer. Return value could be 0 or -1 for default color.
<i>LineType</i> :	Line type. Valid returns are defined in swLineTypes_e . A line type is a combination of a lineStyle and lineWeight.
<i>LineFont</i> :	Value is used for polyline font information. This value can be used as an input to the IDrawingDoc::GetLineFontInfo2 and IDrawinDoc::GetLineFontName2 .
<i>LineWidth</i> :	Integer value defining the line width. Valid width values as defined in swLineWeights_e .
<i>LayerID</i> :	Integer value indicating which layer holds this entity. Obtain the ILayer object by passing this integer value to ILayerMgr::GetLayerById or ILayerMgr::IGetLayerById .
<i>LayerOverride</i> :	Integer with bit flags set to determine which properties, if any, have been overridden with respect to the Layer default properties. If the bit value is set, then the specific property or properties have been overridden. The bit indicators are: color = 0x1, style = 0x2, and width = 0x4. Therefore, if LayerOverride was returned as 3, you know the color and style have been specifically set for this item and may not match the default values associated with this item's layer.
<i>StartPt</i> [3] :	Array of 3 doubles (X,Y,Z) describing the ellipse start point
<i>EndPt</i> [3] :	Array of 3 doubles (X,Y,Z) describing the ellipse end point. If the ellipse is closed, then this will be the same point as the StartPt.
<i>CenterPt</i> [3] :	Array of 3 doubles (X,Y,Z) describing the ellipse center point.
<i>MajorPt</i> [3] :	Array of 3 doubles (X,Y,Z) describing a point on the ellipse and on the major axis.
<i>MinorPt</i> [3] :	Array of 3 doubles (X,Y,Z) describing a point on the ellipse and on the minor axis.

Direction : -1 for clockwise, +1 for counterclockwise

This set of data repeats for each ellipse in the sketch. The size of the array is (NumEllipses * 20).

To determine the number of ellipses, use [ISketch::GetEllipseCount](#).

See Also

[ISketch Interface](#)
[ISketch Members](#)
[ISketch::GetEllipses3 Method](#)
[ISketchEllipse Interface](#)

Availability

SOLIDWORKS 2003 FCS, Revision Number 11.0

SOLIDWORKS API Help

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IGetLines Method (ISketch)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISketch Interface](#) : IGetLines Method (ISketch)

Obsolete. Superseded by [ISketch::GetLines2](#) and [ISketch::IGetLines2](#).

.NET Syntax

Visual Basic (Declaration)

```
Function IGetLines() As System.Double
```

Visual Basic (Usage)



```
Dim instance As ISketch  
Dim value As System.Double  
  
value = instance.IGetLines()
```

C#

```
System.double IGetLines()
```

C++/CLI

```
System.double IGetLines();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Visual Basic for Applications (VBA) Syntax

See [Sketch::IGetLines](#).

See Also

[ISketch Interface](#)
[ISketch Members](#)

SOLIDWORKS API Help

[Send comments](#) on this topic.

IGetLines2 Method (ISketch)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISketch Interface](#) : IGetLines2 Method (ISketch)

Gets all of the lines in the sketch with an option to include or exclude crosshatch lines.

.NET Syntax

Visual Basic (Declaration)

```
Function IGetLines2( _
    ByVal CrossHatchOption As System.Short, _
    ByVal ArraySize As System.Integer _
) As System.Double
```

Visual Basic (Usage)



```
Dim instance As ISketch
Dim CrossHatchOption As System.Short
Dim ArraySize As System.Integer
Dim value As System.Double

value = instance.IGetLines2(CrossHatchOption, ArraySize)
```

C#

```
System.double IGetLines2(
    System.short CrossHatchOption,
    System.int ArraySize
)
```

C++/CLI

```
System.double IGetLines2(
    & System.short CrossHatchOption,
    & System.int ArraySize
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

CrossHatchOption

Crosshatch option as defined in [swCrossHatchFilter_e](#)

ArraySize

Number of lines in the sketch

Return Value

- in-process, unmanaged C++: Pointer to an array of doubles (see **Remarks**)
 - VBA, VB.NET, C#, and C++/CLI: Not supported
- See [In-process Methods](#) for details about this type of method.

Remarks

The return value is the following array of doubles:

[*Color*, *Type*, *Line Font*, *Line Width*, *Layer ID*, *Layer Override*, [*Start*], [*End*]]

where all data values are returned as doubles:

Color = COLORREF returned as an integer. Return value could be 0 or -1 for default color.

Type = line type. Valid returns as defined in [swLineTypes_e](#). A line type is a combination of a line style and line weight.

Line Font = line style. Valid line styles as defined in [swLineStyles_e](#).

Line Width = integer value defining the line width. Valid width values as defined in [swLineWeights_e](#).

Layer ID = integer value indicating which layer holds this entity. Obtain the [ILayer](#) object by passing this integer value to [ILayerMgr::GetLayerById](#) or [ILayerMgr::!GetLayerById](#).

Layer Override = integer with bit flags set to determine which properties, if any, have been overridden with respect to the Layer default properties. If the bit value is set, then the specific property or properties have been overridden. The bit indicators are: color = 0x1, style = 0x2, and width = 0x4. Therefore, if LayerOverride is returned as 3, you know the color and style have been specifically set for this item and may not match the default values associated with this item's layer.

Start[3] = array of 3 doubles (X,Y,Z) describing the line start point.

End[3] = array of 3 doubles (X,Y,Z) describing the line end point.

This set of data repeats for each line in the sketch. The number of doubles returned is (lineCount * 12). To determine the number of lines in the sketch, use [ISketch::GetLineCount2](#).

See [ISketch::GetSketchSegments](#) or [ISketch::!EnumSketchSegments](#) for access to individual [ISketchSegment](#) and [ISketchLine](#) objects.

See Also

[ISketch Interface](#)
[ISketch Members](#)
[ISketch::GetLines2 Method](#)
[ISketchLine Interface](#)

Availability

SOLIDWORKS 2003 FCS, Revision Number 11.0

SOLIDWORKS API Help

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IGetParabolas Method (ISketch)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISketch Interface](#) : IGetParabolas Method (ISketch)

Obsolete. Superseded by [ISketch::GetParabolas2](#) and [ISketch::IGetParabolas2](#).

.NET Syntax

Visual Basic (Declaration)

```
Function IGetParabolas() As System.Double
```

Visual Basic (Usage)



```
Dim instance As ISketch  
Dim value As System.Double  
  
value = instance.IGetParabolas()
```

C#

```
System.double IGetParabolas()
```

C++/CLI

```
System.double IGetParabolas();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Visual Basic for Applications (VBA) Syntax

See [Sketch::IGetParabolas](#).

See Also

[ISketch Interface](#)
[ISketch Members](#)

SOLIDWORKS API Help

[Send comments](#) on this topic.

IGetParabolas2 Method (ISketch)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISketch Interface](#) : IGetParabolas2 Method (ISketch)

Gets all of the parabolas in the sketch.

.NET Syntax

Visual Basic (Declaration)

```
Function IGetParabolas2( _  
    ByVal ArraySize As System.Integer _  
) As System.Double
```

Visual Basic (Usage)



```
Dim instance As ISketch
Dim ArraySize As System.Integer
Dim value As System.Double

value = instance.IGetParabolas2(ArraySize)
```

C#

```
System.double IGetParabolas2(
    System.int ArraySize
)
```

C++/CLI

```
System.double IGetParabolas2(
    & System.int ArraySize
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

ArraySize

Number of parabolas in the sketch

Return Value

- in-process, unmanaged C++: Pointer to an array of doubles (see **Remarks**)
 - VBA, VB.NET, C#, and C++/CLI: Not supported
- See [In-process Methods](#) for details about this type of method.

Remarks

See [ISketch::GetSketchSegments](#) or [ISketch::IEnumSketchSegments](#) for access to individual [ISketchSegment](#) and [ISketchParabola](#) objects.

The return values are in an array of doubles:

[*Color*, *LineType*, *LineStyleIndex*, *LineWidth*, *LayerID*, *LayerOverride*, *StartPt*[3], *EndPt*[3], *FocusPt*[3], *ApexPt*[3]]

where:

Color = COLORREF returned as an integer. Return value could be 0 or -1 for default color.

LineType = line type. Valid returns as defined in [swLineTypes e](#). A line type is a combination of a line style and line weight.

LineStyleIndex = line style. Valid line styles as defined in [swLineStyles e](#).

LineWidth = integer value defining the line width. Valid width values as defined in [swLineWeights e](#).

LayerID = integer value indicating which layer holds this entity. Obtain the [ILayer](#) object by passing this integer value to [ILayerMgr::GetLayerById](#) or [ILayerMgr::IGetLayerById](#).

LayerOverride = integer with bit flags set to determine which properties, if any, have been overridden with respect to the Layer default properties. If the bit value is set, then the specific property or properties have been overridden. The bit indicators are: color = 0x1, style = 0x2, and width = 0x4. Therefore, if LayerOverride was returned as 3, you know the color and style have been specifically set for this item and may not match the default values associated with this item's layer.

StartPt[3] = array of 3 doubles (X,Y,Z) describing the parabola start point.

EndPt[3] = array of 3 doubles (X,Y,Z) describing the parabola end point.

FocusPt[3] = array of 3 doubles (X,Y,Z) describing the parabola focus point.

ApexPt[3] = array of 3 doubles (X,Y,Z) describing the parabola apex point.

This set of data repeats for each parabola in the sketch. The size of the array is (NumParabolas * 18). To determine the number of parabolas, see [ISketch::GetParabolaCount](#).

See Also

[ISketch Interface](#)
[ISketch Members](#)
[ISketch::GetParabolas2 Method](#)
[ISketchParabola Interface](#)

Availability

IGetPolylines Method (ISketch)

[See Also](#)[SolidWorks.Interop.sldworks Namespace](#) > [ISketch Interface](#) : IGetPolylines Method (ISketch)

Not implemented.

.NET Syntax

Visual Basic (Declaration)

```
Function IGetPolylines() As System.Double
```

Visual Basic (Usage)



```
Dim instance As ISketch  
Dim value As System.Double  
  
value = instance.IGetPolylines()
```

C#

```
System.double IGetPolylines()
```

C++/CLI

```
System.double IGetPolylines();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

See Also

[ISketch Interface](#)
[ISketch Members](#)

IGetSketchBlockInstances Method (ISketch)

[See Also](#)[SolidWorks.Interop.sldworks Namespace](#) > [ISketch Interface](#) : IGetSketchBlockInstances Method (ISketch)

Gets the block instances in this sketch (i.e., the sketch under which the block instances are displayed in the FeatureManager design tree).

.NET Syntax

Visual Basic (Declaration)

```
Function IGetSketchBlockInstances( _  
    ByVal Count As System.Integer _  
) As SketchBlockInstance
```

Visual Basic (Usage)



```
Dim instance As ISketch  
Dim Count As System.Integer  
Dim value As SketchBlockInstance  
  
value = instance.IGetSketchBlockInstances(Count)
```

C#

```
SketchBlockInstance IGetSketchBlockInstances(  
    System.int Count  
)
```

C++/CLI

```
SketchBlockInstance^ IGetSketchBlockInstances(  
& System.int Count  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Count

Number of block instances in this sketch

Return Value

- in-process, unmanaged C++: Pointer to an array containing the [block instances](#) in this sketch
 - VBA, VB.NET, C#, and C++/CLI: Not supported
- See [In-process Methods](#) for details about this type of method.

Remarks

Call [ISketch::GetSketchBlockInstanceCount](#) before calling this method to get the number of block instances in this sketch.

See [Block Definitions and Block Instances](#) for details.

See Also

[ISketch Interface](#)
[ISketch Members](#)
[ISketch::GetSketchBlockInstances Method](#)
[ISketchBlockInstance Interface](#)
[ISketchBlockDefinition Interface](#)

Availability

SOLIDWORKS 2007 FCS, Revision Number 15.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

IGetSketchContours Method (ISketch)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISketch Interface](#) : IGetSketchContours Method (ISketch)

Gets the sketch contours in this sketch.

.NET Syntax

Visual Basic (Declaration)

```
Function IGetSketchContours( _  
    ByVal Count As System.Integer _  
) As SketchContour
```

Visual Basic (Usage)



```
Dim instance As ISketch  
Dim Count As System.Integer  
Dim value As SketchContour  
  
value = instance.IGetSketchContours(Count)
```

C#

```
SketchContour IGetSketchContours(  
    System.int Count  
)
```

C++/CLI

```
SketchContour^ IGetSketchContours(  
& System.int Count  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Count
Number of sketch contours

Return Value

- in-process, unmanaged C++: Pointer to an array of [sketch contours](#)
 - VBA, VB.NET, C#, and C++/CLI: Not supported
- See [In-process Methods](#) for details about this type of method.

Remarks

Call [ISketch::GetSketchContourCount](#) before calling this method to get Count.

This method works even if the sketch is suppressed.

See Also

[ISketch Interface](#)
[ISketch Members](#)
[ISketch::GetSketchContours Method](#)

Availability

SOLIDWORKS 2009 FCS, Revision Number 17.0

SOLIDWORKS API Help

[Send comments](#) on this topic.

IGetSketchPaths Method (ISketch)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISketch Interface](#) : IGetSketchPaths Method (ISketch)

Gets the sketch paths in this sketch.

.NET Syntax

Visual Basic (Declaration)

```
Function IGetSketchPaths( _  
    ByVal Count As System.Integer _  
) As SketchPath
```

Visual Basic (Usage)



```
Dim instance As ISketch  
Dim Count As System.Integer  
Dim value As SketchPath  
  
value = instance.IGetSketchPaths(Count)
```

C#

```
SketchPath IGetSketchPaths(  
    System.int Count  
)
```

C++/CLI

```
SketchPath^ IGetSketchPaths(  
& System.int Count  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Count
Number of sketch paths

Return Value

- in-process, unmanaged C++: Pointer to an array of [sketch paths](#)
 - VBA, VB.NET, C#, and C++/CLI: Not supported
- See [In-process Methods](#) for details about this type of method.

Remarks

Call [ISketch::GetSketchPathCount](#) before calling this method to get the value of Count.

See Also

[ISketch Interface](#)
[ISketch Members](#)
[ISketch::GetSketchPaths Method](#)

Availability

SOLIDWORKS 2008 FCS, Revision Number 16.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

IGetSketchPictures Method (ISketch)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISketch Interface](#) : IGetSketchPictures Method (ISketch)

Gets the pictures on this sketch.

.NET Syntax

Visual Basic (Declaration)

```
Function IGetSketchPictures( _  
    ByVal Count As System.Integer _  
) As SketchPicture
```

Visual Basic (Usage)



```
Dim instance As ISketch  
Dim Count As System.Integer  
Dim value As SketchPicture  
  
value = instance.IGetSketchPictures(Count)
```

C#

```
SketchPicture IGetSketchPictures(  
    System.int Count  
)
```

C++/CLI

```
SketchPicture^ IGetSketchPictures(  
& System.int Count  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Count
 Number of pictures on this sketch

Return Value

- in-process, unmanaged C++: Pointer to an array of [pictures](#) on this sketch
 - VBA, VB.NET, C#, and C++/CLI: Not supported
- See [In-process Methods](#) for details about this type of method.

Visual Basic for Applications (VBA) Syntax

See [Sketch::IGetSketchPictures](#).

Remarks

Call [ISketch::GetSketchPictureCount](#) before calling this method to get the value of Count.

See Also

[ISketch Interface](#)
[ISketch Members](#)
[ISketch::GetSketchPictures Method](#)

Availability

SOLIDWORKS 2008 FCS, Revision Number 16.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

IGetSketchPoints2 Method (ISketch)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISketch Interface](#) : IGetSketchPoints2 Method (ISketch)

Gets the sketch points in this sketch.

.NET Syntax

Visual Basic (Declaration)

```
Function IGetSketchPoints2( _  
    ByVal Count As System.Integer _  
) As SketchPoint
```

Visual Basic (Usage)



```
Dim instance As ISketch  
Dim Count As System.Integer  
Dim value As SketchPoint  
  
value = instance.IGetSketchPoints2(Count)
```

C#

```
SketchPoint IGetSketchPoints2(  
    System.int Count  
)
```

C++/CLI

```
SketchPoint^ IGetSketchPoints2(  
& System.int Count  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Count

Number of sketch points in this sketch

Return Value

- in-process, unmanaged C++: Pointer to an array of [sketch points](#) in this sketch
 - VBA, VB.NET, C#, and C++/CLI: Not supported
- See [In-process Methods](#) for details about this type of method.

Remarks

Call [ISketch::GetSketchPointsCount2](#) before calling this method to get the number of sketch points in this sketch.

See Also

[ISketch Interface](#)
[ISketch Members](#)
[ISketch::GetSketchPoints2 Method](#)
[ISketch::GetUserPoints2 Method](#)
[ISketch::GetUserPointsCount Method](#)
[ISketch::IGetUserPoints2 Method](#)
[ISketch::IEnumSketchPoints Method](#)
[ISketchPoint Interface](#)
[ISketch::MergePoints Method](#)

Availability

SOLIDWORKS 2006 SP2, Revision Number 14.2

SOLIDWORKS API Help

[Send comments](#) on this topic.

IGetSketchRegions Method (ISketch)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISketch Interface](#) : IGetSketchRegions Method (ISketch)

Gets the sketch regions in this sketch.

.NET Syntax

Visual Basic (Declaration)

```
Function IGetSketchRegions( _  
    ByVal Count As System.Integer _  
) As SketchRegion
```

Visual Basic (Usage)



```
Dim instance As ISketch  
Dim Count As System.Integer  
Dim value As SketchRegion  
  
value = instance.IGetSketchRegions(Count)
```

C#

```
SketchRegion IGetSketchRegions(  
    System.int Count  
)
```

C++/CLI

```
SketchRegion^ IGetSketchRegions(  
& System.int Count  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Count

Number of sketch regions

Return Value

- in-process, unmanaged C++: Pointer to an array of [sketch regions](#)
 - VBA, VB.NET, C#, and C++/CLI: Not supported
- See [In-process Methods](#) for details about this type of method.

Remarks

Before calling this method, call [ISketch::GetSketchRegionCount](#) to get Count.

See Also

[ISketch Interface](#)
[ISketch Members](#)
[ISketch::GetSketchRegions Method](#)

Availability

IGetSketchSlots Method (ISketch)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISketch Interface](#) : IGetSketchSlots Method (ISketch)

Gets the sketch slots in this sketch.

.NET Syntax

Visual Basic (Declaration)

```
Function IGetSketchSlots( _
    ByVal Count As System.Integer _
) As SketchSlot
```

Visual Basic (Usage)



```
Dim instance As ISketch
Dim Count As System.Integer
Dim value As SketchSlot

value = instance.IGetSketchSlots(Count)
```

C#

```
SketchSlot IGetSketchSlots(
    System.int Count
)
```

C++/CLI

```
SketchSlot^ IGetSketchSlots(
    & System.int Count
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Count

Number of sketch slots in this sketch

Return Value

- in-process, unmanaged C++: Pointer to an array of [sketch slots](#)
 - VBA, VB.NET, C#, and C++/CLI: Not supported
- See [In-process Methods](#) for details about this type of method.

Remarks

Before calling this method, call [ISketch::GetSketchSlotCount](#) to get Count.

See Also

[ISketch Interface](#)
[ISketch Members](#)
[ISketch::GetSketchSlots Method](#)

Availability

SOLIDWORKS 2009 FCS, Revision Number 17.0

IGetSplineParams Method (ISketch)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISketch Interface](#) : IGetSplineParams Method (ISketch)

Obsolete. Superseded by [ISketch::GetSplineParams2](#) and [ISketch::IGetSplineParams2](#).

.NET Syntax

Visual Basic (Declaration)

```
Function IGetSplineParams() As System.Double
```

Visual Basic (Usage)



```
Dim instance As ISketch  
Dim value As System.Double
```

```
value = instance.IGetSplineParams()
```

C#

```
System.double IGetSplineParams()
```

C++/CLI

```
System.double IGetSplineParams();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Visual Basic for Applications (VBA) Syntax

See [Sketch::IGetSplineParams](#).

See Also

[ISketch Interface](#)
[ISketch Members](#)

SOLIDWORKS API Help

[Send comments](#) on this topic.

IGetSplineParams2 Method (ISketch)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISketch Interface](#) : IGetSplineParams2 Method (ISketch)

Obsolete. Superseded by [ISketch::IGetSplineParameters3](#).

.NET Syntax

Visual Basic (Declaration)

```
Function IGetSplineParams2() As System.Double
```

Visual Basic (Usage)



```
Dim instance As ISketch  
Dim value As System.Double
```

```
value = instance.IGetSplineParams2()
```

C#

```
System.double IGetSplineParams2()
```

C++/CLI

```
System.double IGetSplineParams2();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

- in-process, unmanaged C++: Pointer to an array of doubles (see **Remarks**)
- VBA, VB.NET, C#, and C++/CLI: Not supported
See [In-process Methods](#) for details about this type of method.

Remarks

See [ISketch::GetSketchSegments](#) or [ISketch::IEnumSketchSegments](#) for access to individual [ISketchSegment](#) and [ISketchSpline](#) objects. The return value is an array of doubles containing data for all the splines in the sketch:

[*packedDouble1*, *packedDouble2*, *ControlPoint1*[*Dimension elements*], *ControlPoint2*[*Dimension elements*],... *knot1*, *knot2*,..., *packedDouble3*, *packedDouble4*, *packedDouble5*,]

packedDouble1* and *packedDouble2

The first two array elements for each spline contain four integer values holding information that describes the rest of the data in that spline's parameters:

Spline Element	Packed Data	
	low part	high part
0	<i>Dim</i>	<i>Order</i>
1	<i>nCtrlPoints</i>	<i>Periodic</i>

where:

- *Dim* is the number of dimensions in which the spline is defined
- *Order* is the order of the spline
- *nCtrlPoints* is the number of control points
- *Periodic* is 1 for a closed spline or 0 for an open spline

ControlPoint#

The ControlPoint data (in the sketch coordinate system) follows the two packed data elements.

knots#

The number of knots depends on whether the spline is periodic or not:

Periodic:	$\text{numKnots} = \text{nCtrlPoints} + 1$
Non-Periodic:	$\text{numKnots} = \text{nCtrlPoints} + \text{Order}$

packedDouble3*, *packedDouble4*, and *packedDouble5

The last three array elements for each spline contain five integer values holding style and layer information:

Spline Element	Packed Data	
	low part	high part
<i>i</i>	<i>Color</i>	<i>lineStyle</i>
<i>i+1</i>	<i>lineWidth</i>	<i>Layer</i>
<i>i+2</i>	<i>layerOverride</i>	<i>Not used</i>

where:

- *i* is the index following the last Knot or $[2 + \text{numKnots} + \text{numControlPointDoubles} * \text{Dim}]$
- *Color* is the COLORREF value describing the color used for the *ith* spline
- *lineStyle* is the line style used for the *ith* spline. Valid values can be found in the [swLineStyle_e](#) enumeration
- *lineWidth* is line width used for the *ith* spline. Valid values can be found in the [swLineWeights_e](#) enumeration
- *Layer* is an integer index to the layer that the *ith* spline is on
- *layerOverride* is integer with bit flags set to determine which properties, if any, have been overridden or should be overridden.

Therefore, the size of the data for each spline is given by:

$$2 + \text{numKnots} + \text{numControlPointDoubles} * \text{Dim} + 3$$

See Also

[ISketch Interface](#)
[ISketch Members](#)
[ISketch::GetSplineParams2 Method](#)
[ISketch::GetSplineCount Method](#)
[ISketch::GetSplineInterpolateCount Method](#)
[ISketch::GetSplines Method](#)
[ISketch::GetSplinesInterpolate Method](#)
[ISketch::ISplines Method](#)
[ISketch::ISplinesInterpolate Method](#)

Availability

SOLIDWORKS 2000 FCS, Revision Number 8.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

IGetSplineParams3 Method (ISketch)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISketch Interface](#) : IGetSplineParams3 Method (ISketch)

Obsolete. Superseded by [ISketch::GetSplineParams4](#).

.NET Syntax

Visual Basic (Declaration)

```
Function IGetSplineParams3( _  
    ByVal ForceNonPeriodic As System.Boolean _  
) As System.Double
```

Visual Basic (Usage)



```
Dim instance As ISketch  
Dim ForceNonPeriodic As System.Boolean  
Dim value As System.Double  
  
value = instance.IGetSplineParams3(ForceNonPeriodic)
```

C#

```
System.double IGetSplineParams3(  
    System.bool ForceNonPeriodic  
)
```

C++/CLI

```
System.double IGetSplineParams3(  
& System.bool ForceNonPeriodic  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

ForceNonPeriodic

True to attempt to convert all non-periodic splines to periodic, false to not

Return Value

- in-process, unmanaged C++: Pointer to an array of doubles (see **Remarks**)
 - VBA, VB.NET, C#, and C++/CLI: Not supported
- See [In-process Methods](#) for details about this type of method.

Remarks

See [ISketch::GetSketchSegments](#) or [ISketch::IEnumerableSketchSegments](#) for access to individual [ISketchSegment](#) and [ISketchSpline](#) objects.

The return value is an array of doubles containing data for all the splines in the sketch:

[*packedDouble1*, *packedDouble2*, *ControlPoint1*[*Dimension elements*], *ControlPoint2*[*Dimension elements*],... *knot1*, *knot2*,..., *packedDouble3*, *packedDouble4*, *packedDouble5*,]

packedDouble1 and packedDouble2

The first two array elements for each spline contain four integer values holding information that describes the rest of the data in that spline's parameters:

Spline Element	Packed Data	
	low part	high part
0	<i>Dim</i>	<i>Order</i>
1	<i>nCtrlPoints</i>	<i>Periodic</i>

where:

- *Dim* is the number of dimensions in which the spline is defined
- *Order* is the order of the spline
- *nCtrlPoints* is the number of control points
- *Periodic* is 1 for a closed spline or 0 for an open spline

ControlPoint#

The ControlPoint data (in the sketch coordinate system) follows the two packed data elements.

knots#

The number of knots depends on whether the spline is periodic or not:

Periodic:	$\text{numKnots} = \text{nCtrlPoints} + 1$
Non-Periodic:	$\text{numKnots} = \text{nCtrlPoints} + \text{Order}$

packedDouble3, packedDouble4, and packedDouble5

The last three array elements for each spline contain five integer values holding style and layer information:

Spline Element	Packed Data	
	low part	high part
<i>i</i>	<i>Color</i>	<i>lineStyle</i>
<i>i+1</i>	<i>lineWidth</i>	<i>Layer</i>
<i>i+2</i>	<i>layerOverride</i>	<i>Not used</i>

where:

- *i* is the index following the last Knot or $[2 + \text{numKnots} + \text{numControlPointDoubles} * \text{Dim}]$
- *Color* is the COLORREF value describing the color used for the *ith* spline
- *lineStyle* is the line style used for the *ith* spline. Valid values can be found in the [swLineStyles_e](#) enumeration
- *lineWidth* is line width used for the *ith* spline. Valid values can be found in the [swLineWeights_e](#) enumeration
- *Layer* is an integer index to the layer that the *ith* spline is on
- *layerOverride* is integer with bit flags set to determine which properties, if any, have been overridden or should be overridden.

Therefore, the size of the data for each spline is given by:

$$2 + \text{numKnots} + \text{numControlPointDoubles} * \text{Dim} + 3$$

See Also

[ISketch Interface](#)
[ISketch Members](#)
[ISketch::IGetSplines Method](#)
[ISketch::IGetSplinesInterpolate Method](#)
[ISketch::GetSplineParams3 Method](#)
[ISketch::GetSplines Method](#)
[ISketch::GetSplinesInterpolate Method](#)
[ISketch::GetSplineInterpolateCount Method](#)
[ISketch::GetSplineCount Method](#)

Availability

SOLIDWORKS 2010 FCS, Revision Number 18.0

SOLIDWORKS API Help

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IGetSplines Method (ISketch)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISketch Interface](#) : IGetSplines Method (ISketch)

Gets information for each spline by tessellation instead of by interpolation as is done by [ISketch::GetSplinesInterpolate](#) and [ISketch::IGetSplinesInterpolate](#).

.NET Syntax

Visual Basic (Declaration)

```
Function IGetSplines() As System.Double
```

Visual Basic (Usage)



```
Dim instance As ISketch  
Dim value As System.Double  
  
value = instance.IGetSplines()
```

C#

```
System.double IGetSplines()
```

C++/CLI

```
System.double IGetSplines();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

- in-process, unmanaged C++: Pointer to an array of doubles (see **Remarks**)
 - VBA, VB.NET, C#, and C++/CLI: Not supported
- See [In-process Methods](#) for details about this type of method.

Remarks

See [ISketch::GetSketchSegments](#) or [ISketch::IEnumSketchSegments](#) for access to individual [ISketchSegment](#) and [ISketchSpline](#) objects.

The returned array of doubles has the format:

$$\left[\left[\textit{Color}, \textit{LineType}, \textit{NumSplinePoints}, [x,y,z] \right] \right]$$

This complete set of data repeats itself for each spline found in the sketch. For each spline, the array returned contains the color, the line type, the number of spline points in the spline, and the X,Y,Z value for each of those points. Therefore, the [x,y,z] parameter is an array of *NumSplinePoints*, which may vary in size from spline to spline.

The [x,y,z] points for each spline are not the same as the points used to generate the spline. This method tessellates the spline based on the display quality and places points along the spline appropriately. *LineType* may take one of the values in [swLineTypes_e](#).

See Also

[ISketch Interface](#)
[ISketch Members](#)
[ISketch::GetSplines Method](#)

[ISketch::GetSplineCount Method](#)
[ISketch::GetSplineInterpolateCount Method](#)
[ISketch::GetSplineParams2 Method](#)
[ISketch::GetSplineParamsCount2 Method](#)
[ISketch::GetSplinesInterpolate Method](#)
[ISketch::IGetSplineParams2 Method](#)
[ISketch::IGetSplinesInterpolate Method](#)
[ISketchSpline Interface](#)

SOLIDWORKS API Help

[Send comments on this topic.](#)

IGetSplinesInterpolate Method (ISketch)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISketch Interface](#) : IGetSplinesInterpolate Method (ISketch)

Gets the spline points by interpolation instead of by tessellation as is done by [ISketch::GetSplines](#) and [ISketch::IGetSplines](#).

.NET Syntax

Visual Basic (Declaration)

```
Function IGetSplinesInterpolate() As System.Double
```

Visual Basic (Usage)



```
Dim instance As ISketch  
Dim value As System.Double
```

```
value = instance.IGetSplinesInterpolate()
```

C#

```
System.double IGetSplinesInterpolate()
```

C++/CLI

```
System.double IGetSplinesInterpolate();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

- in-process, unmanaged C++: Pointer to an array of doubles (see **Remarks**)
- VBA, VB.NET, C#, and C++/CLI: Not supported
- See [In-process Methods](#) for details about this type of method.

Remarks

To determine the size of the array returned, call [ISketch::GetSplineInterpolateCount](#).

The return value is an array of doubles formatted as:

$$\left[\left[NumSplinePoints, [x, y, z] \right], \right]$$

This complete set of data repeats itself for each spline found in the sketch. For each spline, the array returned contains the number of spline points in the spline, and the X,Y,Z value for each of those points.

The $[x,y,z]$ parameter is an array of *NumSplinePoints*. For example, if your sketch has two splines and each spline has three points, then the data would be in the following format:

$$\left[3, x1_1, y1_1, z1_1, x2_1, y2_1, z2_1, x3_1, y3_1, z3_1, 3, x1_2, y1_2, z1_2, x2_2, y2_2, z2_2, x3_2, y3_2, z3_2 \right]$$

The $[x,y,z]$ points for each spline are the same points used to generate the spline.

See Also

[ISketch Interface](#)
[ISketch Members](#)
[ISketch::GetSplinesInterpolate Method](#)
[ISketch::GetSplineCount Method](#)
[ISketch::GetSplineParams2 Method](#)
[ISketch::GetSplineParamsCount2 Method](#)
[ISketch::IGetSplineParams2 Method](#)

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IGetUserPoints Method (ISketch)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISketch Interface](#) : IGetUserPoints Method (ISketch)

Obsolete. Superseded by [ISketch::GetUserPoints2](#) and [ISketch::IGetUserPoints2](#).

.NET Syntax

Visual Basic (Declaration)

```
Function IGetUserPoints() As System.Double
```

Visual Basic (Usage)



```
Dim instance As ISketch  
Dim value As System.Double  
  
value = instance.IGetUserPoints()
```

C#

```
System.double IGetUserPoints()
```

C++/CLI

```
System.double IGetUserPoints();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Visual Basic for Applications (VBA) Syntax

See [Sketch::IGetUserPoints](#).

See Also

[ISketch Interface](#)
[ISketch Members](#)

SOLIDWORKS API Help

[Send comments](#) on this topic.

IGetUserPoints2 Method (ISketch)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISketch Interface](#) : IGetUserPoints2 Method (ISketch)

Gets all of the user points in this sketch.

.NET Syntax

Visual Basic (Declaration)

```
Function IGetUserPoints2( _  
    ByVal ArraySize As System.Integer _  
) As System.Double
```

Visual Basic (Usage)



```
Dim instance As ISketch  
Dim ArraySize As System.Integer  
Dim value As System.Double  
  
value = instance.IGetUserPoints2(ArraySize)
```

C#

```
System.double IGetUserPoints2(  
    System.int ArraySize  
)
```

C++/CLI

```
System.double IGetUserPoints2(  
& System.int ArraySize  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

ArraySize
Number of user points in the sketch

Return Value

- in-process, unmanaged C++: Pointer to an array of doubles (see **Remarks**)
 - VBA, VB.NET, C#, and C++/CLI: Not supported
- See [In-process Methods](#) for details about this type of method.

Remarks

The return value is an array of doubles with this format:

[*Color*, *Line Font*, *Line Width*, *Layer ID*, *Layer Override*, *ptLoc[3]*]

where:

- *Color* = COLORREF returned as an integer. Return value could be 0 or -1 for default color.
- *Line Font* = line style. Valid line styles can be found in the [swLineStyles_e](#) enumeration.
- *Line Width* = integer value defining the line width. Valid width values can be found in the [swLineWeights_e](#) enumeration.
- *Layer ID* = integer value indicating which layer holds this entity. The [ILayer](#) object can be obtained by passing this integer value to [ILayerMgr::GetLayerById](#) or [ILayerMgr::!GetLayerById](#).
- *Layer Override* = integer with bit flags set to determine which properties, if any, have been overridden with respect to the Layer default properties. If the bit value is set, then the specific property or properties have been overridden. The bit indicators are: color = 0x1, style : 0x2, and width = 0x4. Therefore, if LayerOverride was returned as 3, you know the color and style have been specifically set for this iter and may not match the default values associated with this item's layer.
- *ptLoc[3]* = an array of doubles (x, y, z) describing the point location

This set of data repeats for each user point in the sketch. The size of the array is (NumPts * 8). To determine the number of points in the sketch, use [ISketch::GetUserPointsCount](#).

See [ISketch::GetSketchPoints2](#), [ISketch::!GetSketchPoints2](#), or [ISketch::!EnumSketchPoints](#) for access to individual [ISketchPoint](#) objects.

See Also

[ISketch Interface](#)
[ISketch Members](#)
[ISketch::GetUserPoints2 Method](#)

Availability

SOLIDWORKS 2003 FCS, Revision Number 11.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

IModelToSketchXform Property (ISketch)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISketch Interface](#) : IModelToSketchXform Property (ISketch)

Obsolete. Superseded by [ISketch::ModelToSketchTransform](#).

NOTE: This property is a get-only property. Set is not implemented.

.NET Syntax

Visual Basic (Declaration)

ReadOnly Property IModelToSketchXform As System.Double

Visual Basic (Usage)

```
Dim instance As ISketch  
Dim value As System.Double
```



```
value = instance.IModelToSketchXform
```

C#

```
System.double IModelToSketchXform {get;}
```

C++/CLI

```
property System.double IModelToSketchXform {  
    System.double get();  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Visual Basic for Applications (VBA) Syntax

See [Sketch::IModelToSketchXform](#).

See Also

[ISketch Interface](#)
[ISketch Members](#)

SOLIDWORKS API Help

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InsertRouteLine Method (ISketch)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISketch Interface](#) : InsertRouteLine Method (ISketch)

Inserts a route line in an explode line sketch or a 3D sketch to indicate component relationships.

.NET Syntax

Visual Basic (Declaration)

```
Function InsertRouteLine( _  
    ByVal ItemsToConnect As System.Object, _  
    ByVal Reverse As System.Object, _  
    ByVal AlternatePath As System.Object, _  
    ByVal AlongXYZ As System.Object _  
    ) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As ISketch  
Dim ItemsToConnect As System.Object  
Dim Reverse As System.Object  
Dim AlternatePath As System.Object  
Dim AlongXYZ As System.Object  
Dim value As System.Boolean
```

```
value = instance.InsertRouteLine(ItemsToConnect, Reverse, AlternatePath, AlongXYZ)
```

C#

```
System.bool InsertRouteLine(  
    System.object ItemsToConnect,  
    System.object Reverse,  
    System.object AlternatePath,  
    System.object AlongXYZ  
)
```

C++/CLI

```
System.bool InsertRouteLine(  
& System.Object^ ItemsToConnect,  
& System.Object^ Reverse,  
& System.Object^ AlternatePath,
```

```
& System.Object^ AlongXYZ
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

ItemsToConnect

Array of faces, edges, and vertices to which to connect the route line

Reverse

Array of Booleans indicating whether to reverse the route line at the corresponding item to connect; true to reverse the direction of the route line, false to not

AlternatePath

Array of Booleans indicating whether to display an alternate path at the corresponding item to connect; true to display another possible path for the route line, false to not

AlongXYZ

Array of Booleans indicating whether to create a path parallel to the X, Y, and Z directions from the corresponding item to connect; true to use the X, Y, and Z directions, false to use the shortest route

Return Value

True if a route line is inserted, false if not

Visual Basic for Applications (VBA) Syntax

See [Sketch::InsertRouteLine](#).

Example

[Insert Explode Line Sketch and Route Line \(VB.NET\)](#)

[Insert Explode Line Sketch and Route Line \(VBA\)](#)

[Insert Explode Line Sketch and Route Line \(C#\)](#)

Remarks

You insert a route line in an [explode line sketch](#).

See Also

[ISketch Interface](#)

[ISketch Members](#)

[IAssemblyDoc::AutoExplode Method](#)

Availability

SOLIDWORKS 2010 FCS, Revision Number 18.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

Is3D Method (ISketch)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISketch Interface](#) : Is3D Method (ISketch)

Gets whether this sketch is 2D or 3D.

.NET Syntax

Visual Basic (Declaration)

```
Function Is3D() As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As ISketch
```

```
Dim value As System.Boolean
```

```
value = instance.Is3D()
```

C#

```
System.bool Is3D()
```

C++/CLI

```
System.bool Is3D();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

True if this sketch is 3D, false if it is 2D

Visual Basic for Applications (VBA) Syntax

See [Sketch::Is3D](#).

Example

[Get Sketch Segment and Curve Data \(VBA\)](#)
[Get Whether Sketch Segment is Trimmed \(VBA\)](#)
[Get x,y,z Locations of Points in Sketch \(VBA\)](#)
[Transform Coordinates from Sketch to Model Space \(VBA\)](#)

See Also

[ISketch Interface](#)
[ISketch Members](#)
[Insert3DSketch2 Method](#)

Availability

SOLIDWORKS 99, datecode 1999207

SOLIDWORKS API Help

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IsBoundaryBoxSketch Method (ISketch)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISketch Interface](#) : IsBoundaryBoxSketch Method (ISketch)

Determines whether the sketch is a boundary box.

.NET Syntax

Visual Basic (Declaration)

```
Function IsBoundaryBoxSketch() As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As ISketch  
Dim value As System.Boolean  
  
value = instance.IsBoundaryBoxSketch()
```

C#

```
System.bool IsBoundaryBoxSketch()
```

C++/CLI

```
System.bool IsBoundaryBoxSketch();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

True if the sketch is a boundary box, false if not

Visual Basic for Applications (VBA) Syntax

See [Sketch::IsBoundaryBoxSketch](#).

Example

[Is Selected Feature a Boundary Box Sketch \(C#\)](#)
[Is Selected Feature a Boundary Box Sketch \(VB.NET\)](#)
[Is Selected Feature a Boundary Box Sketch \(VBA\)](#)

See Also

[ISketch Interface](#)
[ISketch Members](#)

Availability

SOLIDWORKS 2011 FCS, Revision Number 19.0

SOLIDWORKS API Help

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IsDerived Method (ISketch)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISketch Interface](#) : IsDerived Method (ISketch)

Gets whether a sketch is derived.

.NET Syntax

Visual Basic (Declaration)

```
Function IsDerived() As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As ISketch  
Dim value As System.Boolean  
  
value = instance.IsDerived()
```

C#

```
System.bool IsDerived()
```

C++/CLI

```
System.bool IsDerived();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

True if the sketch is derived, false if not

Visual Basic for Applications (VBA) Syntax

See [Sketch::IsDerived](#).

Example

[Create Derived or Underived Sketch \(VB.NET\)](#)
[Create Derived or Underived Sketch \(VBA\)](#)
[Create Derived or Underived Sketch \(C#\)](#)

See Also

[ISketch Interface](#)
[ISketch Members](#)
[IModelDoc2::DeriveSketch Method](#)
[IModelDoc2::UnderiveSketch Method](#)

Availability

SOLIDWORKS 2010 FCS, Revision Number 18.0

SOLIDWORKS API Help

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IsShared Method (ISketch)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISketch Interface](#) : IsShared Method (ISketch)

Gets whether this sketch is used by more than one feature.

.NET Syntax

Visual Basic (Declaration)

```
Function IsShared() As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As ISketch  
Dim value As System.Boolean  
  
value = instance.IsShared()
```

C#

```
System.bool IsShared()
```

C++/CLI

```
System.bool IsShared();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

True if this sketch is used by more than one feature, false if not

Visual Basic for Applications (VBA) Syntax

See [Sketch::IsShared](#).

Example

[Determine If Sketch Is Shared \(VBA\)](#)

See Also

[ISketch Interface](#)
[ISketch Members](#)

Availability

SOLIDWORKS 2003 FCS, Revision Number 11.0

SOLIDWORKS API Help

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IsSketchEditable Method (ISketch)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISketch Interface](#) : IsSketchEditable Method (ISketch)

Gets whether this sketch is editable.

.NET Syntax

Visual Basic (Declaration)

```
Function IsSketchEditable() As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As ISketch  
Dim value As System.Boolean
```

```
value = instance.IsSketchEditable()
```

C#

```
System.bool IsSketchEditable()
```

C++/CLI

```
System.bool IsSketchEditable();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

True if this sketch is editable, false if not

Visual Basic for Applications (VBA) Syntax

See [Sketch::IsSketchEditable](#).

See Also

[ISketch Interface](#)
[ISketch Members](#)
[ISketch::SetSketchEditable Method \(\)](#)

Availability

SOLIDWORKS 2020 FCS, Revision Number 28.0